MANIFOLD-AWARE TUNING

WHITE PAPER · 2026

A METHODOLOGY FOR CAPITAL EQUIPMENT

The Self-Tuning Machine

A physics-grounded methodology that makes a complex, many-knobbed machine tune, diagnose, and hold itself on-spec from an operator's plain-language intent — proven end to end on a simulated multi-stage beamline, ready to apply to yours.

A machine-agnostic method · **Autonomous fault isolation** · **Physics-grounded & auditable**

Prepared for **capital-equipment OEMs & their customers**
An introduction to autonomous tuning as a methodology

Marketing white paper

EXECUTIVE SUMMARY

Semiconductor capital equipment has never been more valuable — or harder to keep on-spec. The tools that define a modern process are exquisitely sensitive, densely instrumented, and increasingly operated by teams stretched thin by a global shortage of senior process talent.

This is a methodology for making precision machines self-tuning. It is not a product bolted to one tool, and it does not depend on our hardware or data: **you bring the machine and a physics-grounded model of it, and the methodology brings the approach** that turns it into a machine that tunes and diagnoses itself. An operator states an outcome in plain language — *"bring the output to target," "something has drifted, find the fault," "hold the line within tolerance"* — and the method plans, evaluates its options against that model, and executes the correction through the tool's own controls. Every move is captured as an auditable machine-state record.

The effect is a step-change in how a tool is run: **faster time-to-spec, fewer expert-hours per tool, and diagnosis that is consistent across every machine and every shift.** We have proven the method end to end on a simulated multi-stage beamline; the same approach applies to any precision tool with a physics model and many interacting controls.

THE PROBLEM

Every hour off-spec is measured in millions

$133B
2025 semiconductor equipment sales — a record, on track to $156B by 2027[1]

$1–3M
Cost per hour of downtime on a leading-edge tool; up to $8–10M at the bleeding edge[2]

40–70%
Of tool-down events are handled as unplanned "firefighting," not planned work[3]

+20%
First-time-right gains of up to 20% reported for AI-guided tools in published use cases[2]

Tuning still rides on scarce experts

Bringing a tool to spec, and isolating what drifted when it falls off, still depends on a handful of senior engineers and hard-won tribal knowledge. That expertise is retiring faster than it is being replaced, and the industry-wide talent gap makes every expert-hour more expensive and harder to schedule.

Manual tuning is slow and inconsistent

Corrections vary with the operator, the shift, and the individual machine. Small differences in how tools are tuned surface later as matching and yield problems — and every hour a tool spends being coaxed back to spec is productive capacity a fab has already paid for but cannot use.

> **The bottleneck is no longer the physics of the machine. It is the human bandwidth required to keep the machine at its best.**

THE METHOD

Intent in. On-spec out.

The method sits between the person who knows *what* the machine should do and the machine that knows *how*. The operator works entirely in outcomes. The method does the translation — turning intent into a precise, verified sequence of machine actions, and turning raw machine state back into a plain-language explanation of what it did and why.

OPERATOR'S VIEW · PLAIN LANGUAGE

Something drifted — find the fault and bring the output back to target.

Two faults are stacking. I've isolated the dominant one, corrected it, and verified the line is back inside tolerance.

TRANSLATE

MACHINE'S VIEW · STRUCTURED & LOGGED

```
DIAGNOSE
state  → fault localized
ADJUST
state  → corrected
VERIFY
state  → within tolerance
```

Illustrative. Action labels are generic workflow steps — the method's diagnostic and control internals are not shown.

Speaks the operator's language

Goals, not procedures. A junior operator can run a tool the way a veteran would — no macro to memorize, no knob sequence to recall.

Thinks before it touches hardware

Options are weighed against a physics-grounded model of the machine first, so the tool moves only when the method has a plan it can defend.

Leaves an auditable trail

Every action is a structured, logged state transition — a complete, reviewable record of what changed, when, and why.

CAPABILITIES

Five things a great operator does — automated

This is not a dashboard and not a script. It is a method that carries out the judgment-heavy parts of running a precision tool, end to end.

01

Plain-language operation

Operators express what they want in ordinary terms and the method handles the rest. The skill barrier to running a complex tool drops sharply, and the knowledge of your best engineers becomes available on every shift.

02

Autonomous fault localization

When a tool drifts, the method detects it, isolates the source, and — when several faults stack at once — separates them and ranks each by its impact, so the dominant problem is fixed first. Corrections are applied on operator approval.

03

Feasibility-aware targeting

The method honors the machine's real operating limits. When a request cannot be fully satisfied, it says so plainly and delivers the best achievable point with an honest margin — never a silent miss dressed up as success.

04

Guided optimization

From a cold start or a drifted state, the method finds an efficient path back to an on-spec operating point in as few moves as possible — shrinking the time and material spent getting a tool productive.

05

Safety & governance, by design

Ambiguous or unsafe requests are met with a clarifying question, not a guess. Physical changes pass through a human approval gate. And because every step is a logged state transition, the whole operation is reviewable after the fact — the accountability a fab environment demands.

HUMAN STAYS IN COMMAND

- Clarifies before acting on unclear intent
- Approval gate on any hardware change
- Full, replayable action history

> **It does the reasoning a senior engineer does — consistently, on every tool, at every hour — and shows its work.**

HOW IT WORKS

A model of the machine — not just a model of language

What separates this method from a generic chat assistant is that it reasons against the physics of the tool. Three layers turn an operator's words into a safe, verified change on the hardware.

1

INTENT LAYER

Understand the goal

Plain-language requests become precise objectives and constraints — the target, the tolerances, and the limits that must be respected. When intent is unclear or unsafe, the layer asks rather than assumes.

2

REASONING LAYER

Plan, simulate, verify

Candidate actions are evaluated against a physics-grounded model of the machine *before* anything is committed to hardware, then confirmed against real readings afterward. This is where drift is localized, feasibility is judged, and the shortest path to spec is chosen.

3

MACHINE LAYER

Act through the tool's own controls

Approved changes are executed through the equipment's existing control interface, and each one is written to the record as a structured state transition — the same trail the operator sees translated back into plain language.

THE ENGAGEMENT

What you bring, what we bring

You bring the machine & its model

You supply the equipment and a physics-grounded model or twin of it. We evaluate that model and fit the method to it — no proprietary hardware or dataset of ours is required.

We bring the method & expertise

The manifold-aware modeling, the closed-loop tuning, and the hands-on work to stand it up on your machine and hand it to your operators.

Proven first, then embedded

Validated against a simulator so nothing is at risk, then embedded into your host software through an open agent protocol (MCP).

BUSINESS IMPACT

Where the value lands

Autonomous tuning maps directly onto the economics fabs and OEMs already track. Each capability pulls a lever that shows up in utilization, uptime, and the productivity of scarce talent.

CAPABILITY	OPERATIONAL LEVER	ECONOMIC EFFECT
Guided tuning & faster time-to-spec	More of each tool's hours are productive rather than spent being coaxed to spec	Reliability and availability gains can unlock **>10% latent capacity** without adding tools[4]
Autonomous fault localization	The right fix is identified the first time, shortening time-to-repair	AI-guided diagnosis has improved first-time-right by **up to 20%** in published use cases[2]
Plain-language operation	Expert judgment is available on every shift; fewer escalations to scarce engineers	Scales a workforce constrained by a persistent talent shortage[5]
Consistent, auditable corrections	Tuning no longer varies by operator, shift, or individual machine	Tighter tool-to-tool matching and fewer downstream yield surprises
Feasibility-aware targeting	Operators learn immediately when a target is unreachable, with the best safe alternative	Less wasted material and fewer dead-end tuning attempts

WHERE IT APPLIES FIRST

Charged-particle & beamline tools

The method's first proof is precision beamline equipment — ion implantation and related charged-particle process tools — where multi-parameter tuning, drift, and fault isolation are daily realities and the cost of getting them wrong is high. We have demonstrated it there, end to end, on a physics-grounded simulator.

The ion-implant segment alone is a multi-billion-dollar market, forecast to approach **$5B by 2032**,[6] sitting inside a record wafer-fab-equipment cycle.

WHERE IT GOES NEXT

Any tool with a model and many knobs

The same method generalizes to other precision capital equipment where behavior can be modeled and outcomes depend on tuning many interacting controls. For an OEM, this is a capability you can apply across a portfolio — not a one-tool feature.

THE ONE-LINE CASE

Turn a tool that requires an expert into a tool that contains one.

WHY NOW

Three curves are converging

Autonomous tuning is arriving at exactly the moment the industry needs it and the technology is ready to deliver it.

Record capital intensity

Equipment sales are at all-time highs and climbing toward $156B by 2027.[1] Every tool is more expensive and more precious, so every idle or off-spec hour costs more than ever.

Agentic AI hits the fab floor

AI agents that plan and act — not just answer — are moving into semiconductor manufacturing, now a headline theme at the industry's major venues.[7] The category is being established this year.

The talent gap widens

A structural shortage of experienced process engineers means the expertise to tune and troubleshoot tools is scarcer each year.[5] Software that carries that expertise is no longer optional.

> **Record-value tools, a proven way for AI agents to operate them, and too few experts to go around. This methodology sits precisely at that intersection.**

Bring us your machine

The method can be demonstrated today on our reference beamline simulator — then applied to your tool. You provide the equipment and a physics model; we bring the method and stand up autonomous tuning, live fault isolation, and the full auditable trail with your team.

Start a pilot → hello@[your-domain]

Sources. 1. SEMI, *Year-End Total Semiconductor Equipment Forecast* — equipment sales $133B (2025) rising to $156B (2027); WFE $115.7B (2025). 2. Downtime economics: leading-edge fab-tool downtime is widely estimated at $1–3M/hr (higher at the bleeding edge) across semiconductor field-service analyses. First-time-right: published field-service AI case studies report first-time-fix improvements of up to ~20%. 3. McKinsey & Company, *Need to boost semiconductor fab efficiency? Look to maintenance* — 40–70% of tool-down events handled as firefighting. 4. McKinsey (same) — reliability gains can unlock >10% latent capacity without adding tools. 5. Accenture / SEMI industry analyses on the semiconductor talent shortage. 6. Market-research consensus on the ion-implant / ion-implantation systems segment (~$5B by 2032). 7. SEMI, *Agentic AI for Next-Generation Semiconductor Manufacturing*; SEMICON West 2026 Smart Manufacturing program.

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